

ME 333
MECHANICS OF MATERIALS

Lecture 26: Thick walled Cylinders & Rotating Disks

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Thick-walled Cylinders

Summary :

- The stress field description for thick-walled cylinders subjected to internal and external pressures is:

$$\sigma_{rr} = \frac{P_i R_i^2 - P_o R_o^2}{(R_o^2 - R_i^2)} + \frac{R_i^2 R_o^2 (P_o - P_i)}{r^2 (R_o^2 - R_i^2)}; \quad \sigma_{\theta\theta} = \frac{P_i R_i^2 - P_o R_o^2}{(R_o^2 - R_i^2)} - \frac{R_i^2 R_o^2 (P_o - P_i)}{r^2 (R_o^2 - R_i^2)}$$

- The stresses in plane do not depend upon elastic constants.
- In case of plane stress $\sigma_{zz} = 0$. Then:

$$\epsilon_{zz} = -\frac{\nu}{E} (\sigma_{rr} + \sigma_{\theta\theta}) = -\frac{\nu}{E} (\sigma_{xx} + \sigma_{yy}) = -\frac{2\nu}{E} \frac{(P_o R_o^2 - P_i R_i^2)}{(R_o^2 - R_i^2)}$$

- In case of plane strain $\epsilon_{zz} = 0$. Then:

$$\sigma_{zz} = \nu (\sigma_{rr} + \sigma_{\theta\theta}) = \nu (\sigma_{xx} + \sigma_{yy}) = 2\nu \frac{(P_i R_i^2 - P_o R_o^2)}{(R_o^2 - R_i^2)}$$

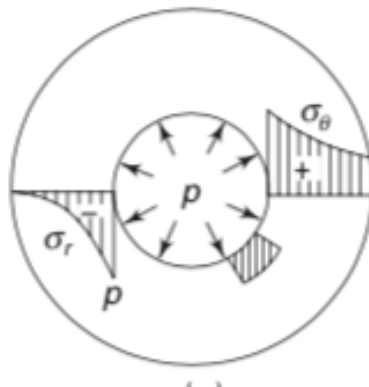
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Thick-walled Cylinders

Stress Profiles : In case of internal pressure only



Ref Base Fig: Advanced Mechanics of Solids, L. S. Srinath

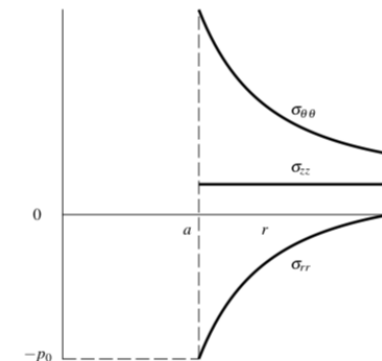
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Thick-walled Cylinders

Stress Profiles : In a general case :



Ref Base Fig: Intermediate Mechanics of Materials, J. R. Barber

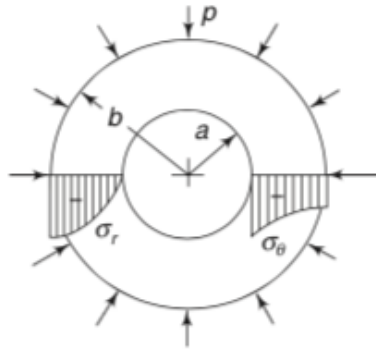
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Thick-walled Cylinders

Stress Profiles : In case of external pressure only



Ref Base Fig: Advanced Mechanics of Solids, L. S. Srinath

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Thick-walled Cylinders

Example : Consider if there is no hole at the origin. Then there exists a material point even at $r = 0$.

- Using general solution :

$$\sigma_{rr} = C_1(1 + 2 \ln r) + 2C_2 + C_3 \frac{1}{r^2}$$

$$\sigma_{\theta\theta} = C_1(3 + 2 \ln r) + 2C_2 + C_3 \frac{1}{r^2}$$

$$\sigma_{r\theta} = 0$$

- In this case, $C_1 = 0$ and $C_3 = 0$. It results in $\sigma_{rr} = \sigma_{\theta\theta} = 2C_2$.
- Assuming no body forces - it is only possible to have a constant state of stress.

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Thick-walled Cylinders

Example 1 : Specialize the stress field description to that for thin-walled cylinders where: $\sigma_{rr} = 0$ and $\sigma_{\theta\theta} = Pr/t$.

- In case of thin walled cylinders, we can write:

$$R_o = R_i \approx r; \quad (R_o - R_i) = t; \quad P_o = 0; \quad P_i = P$$

- The stress field then becomes:

$$\sigma_{rr} = \frac{P_i R_i^2 - P_o R_o^2}{(R_o^2 - R_i^2)} + \frac{R_i^2 R_o^2 (P_o - P_i)}{r^2 (R_o^2 - R_i^2)} = \frac{Pr^2}{2rt} + \frac{r^4 (-P)}{r^2 (2rt)} = 0$$

$$\sigma_{\theta\theta} = \frac{P_i R_i^2 - P_o R_o^2}{(R_o^2 - R_i^2)} - \frac{R_i^2 R_o^2 (P_o - P_i)}{r^2 (R_o^2 - R_i^2)} = \frac{Pr^2}{2rt} - \frac{r^4 (-P)}{r^2 (2rt)} = \frac{Pr}{t}$$

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Thick-walled Cylinders

Example 2 : Determine the radial and hoop stresses at $r = 12.5$ cm in a thick walled cylinder subjected to an internal pressure of 12 MPa. Given $R_i = 10$ cm and $R_o = 15$ cm.

- In this case, we can write:

$$R_o = 15 \text{ cm}; \quad R_i = 10 \text{ cm}; \quad r = 12.5 \text{ cm}; \quad P_o = 0; \quad P_i = 12 \text{ MPa}$$

- The stress field then becomes:

$$\sigma_{rr} = \frac{P_i R_i^2 - P_o R_o^2}{(R_o^2 - R_i^2)} + \frac{R_i^2 R_o^2 (P_o - P_i)}{r^2 (R_o^2 - R_i^2)} = ?$$

$$\sigma_{\theta\theta} = \frac{P_i R_i^2 - P_o R_o^2}{(R_o^2 - R_i^2)} - \frac{R_i^2 R_o^2 (P_o - P_i)}{r^2 (R_o^2 - R_i^2)} = ?$$

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Thick-walled Cylinders

Example 3 : A steel cylinder has $d_i = 1$ m and $P_i = 8$ MPa. Calculate the wall thickness if the maximum shearing stress is not to exceed 35 MPa.

Rotating Disks

- The centripetal acceleration $-\omega^2 r$ acts in the radial direction.
- The equilibrium equation therefore can be written as:

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} = \rho a_r = -\rho \omega^2 r$$

$$\Rightarrow r \frac{\partial \sigma_{rr}}{\partial r} + \sigma_{rr} - \sigma_{\theta\theta} + \rho \omega^2 r^2 = 0$$

- The above equation cannot be solved from equilibrium considerations since it is statically indeterminate.
- **Note:** Axisymmetry ensures that $\sigma_{r\theta} = 0$ and $\frac{\partial \sigma_{\theta\theta}}{\partial \theta} = 0$ implies $\sigma_{\theta\theta}$ is a function of r .

Rotating Disks

- **Step 1 :** The strain-displacement relations are:

$$\epsilon_{rr} = \frac{\partial u_r}{\partial r}; \quad \epsilon_{\theta\theta} = \frac{u_r}{r}$$

- **Step 2 :** Using Hooke's Law (assuming plane stress) :

$$\sigma_{rr} = \frac{E}{(1-\nu^2)} \left(\frac{du_r}{dr} + \nu \frac{u_r}{r} \right); \quad \sigma_{\theta\theta} = \frac{E}{(1-\nu^2)} \left(\frac{u_r}{r} + \nu \frac{du_r}{dr} \right)$$

- **Step 3 :** Substituting back into the equilibrium equation we have :

$$\frac{d}{dr} \left(r \frac{du_r}{dr} + \nu u_r \right) - \nu \frac{du_r}{dr} - \frac{u_r}{r} = -\frac{1-\nu^2}{E} \rho \omega^2 r^2$$

$$\therefore r^2 \frac{d^2 u}{dr^2} + r \frac{du}{dr} - u = -\frac{1-\nu^2}{E} \rho \omega^2 r^3$$

Rotating Disks

- **Step 4 :** Let $u = r^\alpha$. Plugging into the differential equation results in:

$$\alpha(\alpha-1) + \alpha - 1 = 0 \Rightarrow \alpha = \pm 1$$

$$\therefore u = Ar + \frac{B}{r} - \frac{1-\nu^2}{8E} \rho \omega^2 r^3$$

- **Step 5 :** Then the stresses can be written as:

$$\sigma_{rr} = \frac{E}{(1-\nu^2)} \left(A - \frac{B}{r^2} + \nu A + \nu \frac{B}{r^2} \right) - \frac{3+\nu}{8} \rho \omega^2 r^2$$

$$\sigma_{\theta\theta} = \frac{E}{(1-\nu^2)} \left(A - \frac{B}{r^2} + \nu A + \nu \frac{B}{r^2} \right) - \frac{1+3\nu}{8} \rho \omega^2 r^2$$

- **Step 6 :** Apply boundary conditions to determine the unknown coefficients. In case of a solid disk: (to avoid a singularity at $r = 0$) we consider $B = 0$. If 'b' is the radius $\sigma_{rr}(b) = 0$

Rotating Disks

- Step 6 : Contd ...

$$\frac{E}{1-\nu}A = \frac{3+\nu}{8}\rho\omega^2b^2 \implies A = \frac{(3+\nu)(1-\nu)}{8E}\rho\omega^2b^2$$

- Step 7 : Then the stresses can be written as:

$$\sigma_{rr} = \frac{3+\nu}{8}\rho\omega^2(b^2 - r^2)$$

$$\sigma_{\theta\theta} = \frac{(3+\nu)}{8}\rho\omega^2b^2 - \frac{1+3\nu}{8}\rho\omega^2r^2$$

Note : If $\sigma_{rr} = 0$ at the inner radius the constants can be calculated for a ring too.

Rotating Disks

Example 4 : A Rolls-Royce triple shaft turbo used in B-757 has diameter $d = 1$ m, density $\rho = 6500$ kg/m³, $E = 500$ GPa, $\nu = 0.3$ and $\sigma_Y = 600$ MPa. Calculate the max. turbine rotation velocity at which the material first yields using Von-Mises criterion.

Rotating Disks

Q How to compute the max. turbine rotation velocity at which the material first yields ?

- The effective stress measure using Von-Mises criterion is:

$$\sigma_{\text{eff}} = \frac{1}{\sqrt{2}} \sqrt{(\sigma_{rr} - \sigma_{\theta\theta})^2 + \sigma_{\theta\theta}^2 + \sigma_{rr}^2} \quad @ r = 0$$

$$= \frac{1}{\sqrt{2}} \sqrt{2 \left(\frac{(3+\nu)}{8} \rho \omega^2 b^2 \right)^2}$$

$$= \frac{(3+\nu)}{8} \rho \omega^2 b^2 = \sigma_Y$$

- Maximum angular velocity can therefore be calculated as:

$$\omega_{\text{max}} = \sqrt{\frac{8\sigma_{\text{eff}}}{(3+\nu)\rho b^2}} = \sqrt{\frac{8\sigma_Y}{(3+\nu)\rho b^2}}$$

Rotating Disks

Example 5 : An elastic disk is pressfit on to a shaft with an overlap δ . Given: $R_i = 0.02$ m, $R_o = 0.2$ m, $h = 0.04$ m, $\delta = 0.02 \times 10^{-3}$ m, $\rho = 7800$ kg/m³, $E = 200$ GPa and $\nu = 0.3$. How large can the maximum angular velocity be if we want the disc not to get loose?

Rotating Disks

Q How large can the maximum angular velocity be if we want the disc not to get loose?

- The radial displacement at the inner radius:

$$u(R_i) = \delta = AR_i + \frac{B}{R_i} + CR_i^3 = AR_i + \frac{B}{R_i} - \frac{1-\nu^2}{8E} \rho \omega^2 R_i^3$$

- The stress field at the inner and outer radii at the onset of the elastic disk coming loose:

$$\sigma_{rr}|_{(R_i)} = 0 \implies \frac{E}{1-\nu} A - \frac{E}{1+\nu} \frac{B}{R_i^2} - \frac{3+\nu}{8} \rho \omega^2 R_i^2 = 0$$

$$\sigma_{rr}|_{(R_0)} = 0 \implies \frac{E}{1-\nu} A - \frac{E}{1+\nu} \frac{B}{R_0^2} - \frac{3+\nu}{8} \rho \omega^2 R_0^2 = 0$$

Solving for A and B from above.

Rotating Disks

Q How large can the maximum angular velocity be if we want the disc not to get loose?

- Constants A and B take the values:

$$\frac{E}{1-\nu} A = \frac{3+\nu}{8} \rho \omega^2 (R_0^2 + R_i^2)$$

$$\frac{E}{1+\nu} B = \frac{3+\nu}{8} \rho \omega^2 R_0^2 R_i^2$$

- Substituting into the displacement field we get:

$$\delta = \frac{(3+\nu)(1-\nu)}{8E} \rho \omega^2 (R_0^2 + R_i^2) R_i + \frac{(3+\nu)(1+\nu)}{8E} \rho \omega^2 \left(\frac{R_0^2 R_i^2}{R_i} \right) - \frac{1-\nu^2}{8E} \rho \omega^2 R_i^3$$

From here, with given δ we can compute ω .

Reading Assignment

Rework the Example Problems